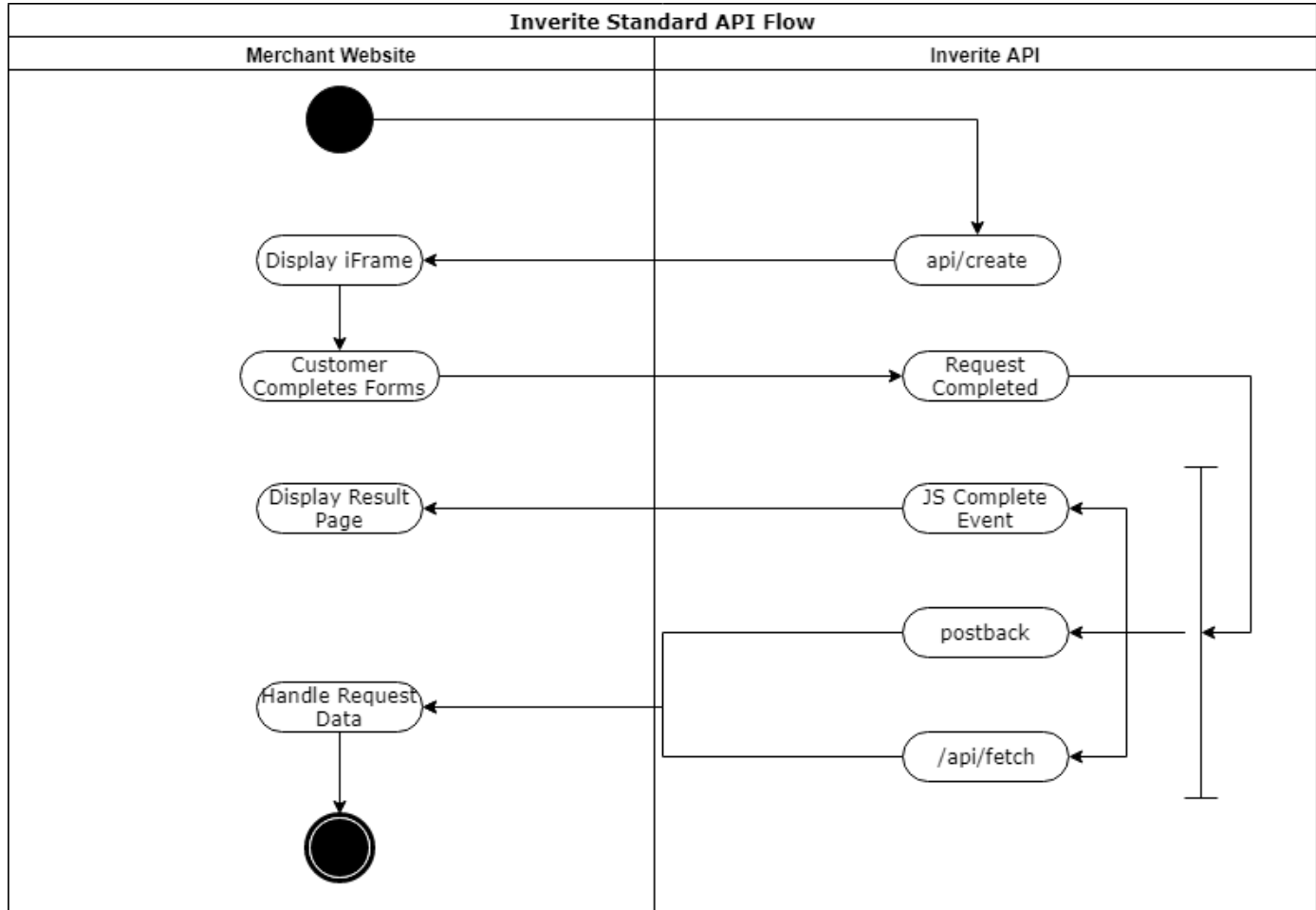


For the purposes of this document, all URLs will reference Sandbox for merchant integration. These URLs change when moving to the live environment. Please note, **all siteID and bankID values are different** between these two systems. Likewise, a GUID is unique to either Sandbox, or your Live account.

All API hits require an Authentication Header to be submitted ("Auth"), with your merchant key. Again, please note this key changes between sandbox and live environment. All APIs will respond in JSON format, with examples included below.



The first API that should be used is the List API. This allows your developer to determine the configured Site information in your account, as well as supported Banks if required. If you need additional sites setup, please contact merchantsupport@inverite.com

API Endpoint URL: <https://sandbox.inverite.com/api/v2/list>

Parameters

<u>type</u>	<u>string ("banks" or "sites")</u>	<u>"banks" to return a list of supported banks;</u> <u>"sites" to return a list configured Merchant sites</u>
-------------	------------------------------------	--

Response

<u>sites</u>	<u>array(siteID, url, displayname)</u>	<u>List of Configured Sites</u>
<u>banks</u>	<u>array(bankID, displayname)</u>	<u>List of Supported Banks</u>

Example List Post

```
curl -v -H "Accept: application/json" \  
      -H "Content-type: application/json" \  
      -H "Auth: your-secret-key" \  
      -X POST https://sandbox.inverite.com/api/v2/list \  
      -d '{"type":"sites"}
```

Example Response (please note, your siteID will differ from "9", please use the List API to determine)

```
{"sites":[{"siteID":"9","url":"https://yourfinancialsite.com/","displayname":"Example Site"]}}
```

In order to create a request, and optionally send it to your customer via Email or Text, the Create API should be used.

Depending on your use case, you may choose to send either the “email” or the “username” parameter. Both should not be used. If you wish for Inverite to email the request to your customer directly, please include the email parameter, and the customers valid email address. If you wish to use Inverite anonymously, or not have us dispatch an email, you should use the username parameter. Both cases should be paired with sending a “referenceid” of the customers unique identifier in your own database. This is used to tie multiple requests together for a single customer. We recommend setting a username of “SiteName_(your internal customerID)” for ease of use. The returned request_guid should be stored in your customer database later association of data.

API Endpoint URL: <https://sandbox.inverite.com/api/v2/create>

Parameters

<u>Field</u>	<u>Required</u>	<u>Description</u>
email	1 *	Customer Email Address
username	1 *	Username (if no email provided). Recommend: YourSiteName_(referenceid)
phone	0	Phone Number to text Verification Request to. Ex 6045551313
ip	0	Customer IP address
siteID	1	Site identifier
requestedfields	0	JSON Array; allowed values: transactions,name,address,accounts
referenceid	1	Unique customer identifier from Merchant database or CRM.
url	0	URL to pass to customer for iframe method
urlbody	0	URL description to pass to customer for iframe method
firstname	0	Customer First Name
lastname	0	Customer Last Name
type	0	detected Customer device type
bankID	0	only allow specified bankID
province	0	Require Provincial regulatory compliance

* email **OR** username are required.

Response

<u>Field</u>	<u>Required</u>	<u>Description</u>
--------------	-----------------	--------------------

request_guid	1	Request GUID (should be stored in your database)
iframeurl	1	URL to redirect user for Iframe type verification
username	1	Created Inverite username for customer
password	0	On initial request will return password for Inverite user for customer
message	0	Error message

Example Create Post

```
curl -v -H "Accept: application/json" \
      -H "Content-type: application/json" \
      -H "Auth: your-secret-key" \
      -X POST https://sandbox.inverite.com/api/v2/create \
      -d
'{"ip":"123.123.123.123","email":"test@inverite.com","firstname":"First","lastname":"Last","siteID":"9","referenceid":"123456","type":"web","days":60}'
```

Example Response

```
{"iframeurl":"https://sandbox.inverite.com/customer/v2/web/start/92334464-513E-11E7-A970-6EC618044181","request_guid":"9633F964-513E-11E7-A970-6EC618044181","username":"test@inverite.com"}
```

Status API

After creating a request, you may wish to query the status. This is a simple query that may be performed for one or more GUID at a time. It returns a list of the guid, and the current status (example "Verified" or "Not Started").

API Endpoint URL: <https://sandbox.inverite.com/api/v2/status>

Parameters

Field	Required	Description
type	1	allowed value: "guid"
guid	1	Array of GUID (string)

Response

<u>status</u>	Array of Objects containing "guid" and "result" string indicating current status
---------------	--

Example Status Post

```
curl -v -H "Accept: application/json" \
  -H "Content-type: application/json" \
  -H "Auth: your-secret-key" \
  -X POST https://sandbox.inverite.com/api/v2/status \
  -d '{"type":"guid","guid":["9633F964-513E-11E7-A970-6EC618044181"]}'
```

Example Response

```
{"status":[{"guid":"9633F964-513E-11E7-A970-6EC618044181","result":"Verified"}]}
```



Fetch API

After a request is completed, you will use the Fetch API to return the full dataset.

API Endpoint URL: <https://sandbox.inverite.com/api/v2/fetch/<GUID>>

Response

complete_datetime	Date and time the verification was completed
request	Guid of the verification request
accounts	Array of Customer financial accounts
contacts	Array of Customer contact information
status	Current status of Request. Will show "Verified"
error	Only Populated if Request not Verified
name	Account holder name(s) from profile
address	Account holder's address from profile

Accounts Array

transit	Transit of financial institution
account	Customer account number
bank	Bank name
institution	Bank institution number
statistics	Dictionary of account metrics
transactions	Array of recent transactions
available_balance	The available balance including overdraft, not always shown
type	"Chequing" or "savings" when positively identified
payschedules	Array of Income dictionaries

Income Dictionary

frequency	payroll frequency and parity
payments	Array of recent payroll transactions

account	deposit account
income_type	monthly, government, self-employed etc
score	confidence indicator
monthly_income	calculated total monthly income
future_payments	array of dates for next payroll dates
details	deposit descriptor

Contacts Array

type	email or phone
contact	phone number or email address

Statistics Dictionary

See "Transaction Insights" document for the complete list of statistics keys.

<u>mean_closing_balance</u>	<u>Average closing balance over verification duration</u>
<u>mean_closing_balance_30</u>	<u>Average closing balance over last 30 days</u>

Transactions Array

<u>balance</u>	<u>Account balance after</u>
<u>category</u>	<u>Transaction category</u>
<u>credit</u>	<u>Amount account was credited</u>
<u>debit</u>	<u>Amount account was debited</u>
<u>date</u>	<u>Date of transaction</u>
<u>details</u>	<u>Transaction details description</u>
<u>flags</u>	<u>"is_regular"</u>

Example Fetch Post

```
curl -v -H "Accept: application/json" \
  -H "Content-type: application/json" \
  -H "Auth: your-secret-key" \
```

-X GET <https://sandbox.inverite.com/api/v2/fetch/9633F964-513E-11E7-A970-6EC618044181>

Example Response

```
{
  "complete_datetime" : "2017-06-14 20:32:24",
  "request" : "9633F964-513E-11E7-A970-6EC618044181",
  "contacts" : [
    {
      "contact" : "johndoe@gmail.com",
      "type" : "email"
    },
    {
      "type" : "phone",
      "contact" : "(604) 555-1234"
    }
  ],
  "accounts" : [
    {
      "bank" : "Inverite",
      "institution" : "001",
      "transit" : "12345",
      "routing_code" : null,
      "account" : "1234567",
      "statistics" : {
        "mean_closing_balance_30" : "221.00",
        "mean_closing_balance" : "163.67"
      },
      "transactions" : [
        {
          "flags" : [
            "is_regular"
          ],
          "category" : "transfer",
          "credit" : "194.71",
          "debit" : "",
          "date" : "2017-06-14",
          "details" : "e-Transfer In Other Reference # 228803",
          "balance" : "217.53"
        }
      ]
    },
    {
      "type" : "chequing",
      "institution" : "123",
      "account" : "1234567",
      "transit" : "12345",
      "bank" : "Test Bank",
      "payschedules" : [
        {
          "frequency" : "bi_weekly:parity_0:thursday",
          "payments" : [
            {
              "details" : "Payroll ACME JOB CO",
              "balance" : "865.82",
              "credit" : "863.92",
              "debit" : "",
              "category" : ""
            }
          ]
        }
      ]
    }
  ]
}
```

```
{
  "flags": [
    "is_payroll"
  ],
  "date": "2020-01-30"
},
{
  "details": "Payroll ACME JOB CO",
  "balance": "985.49",
  "credit": "985.30",
  "date": "2020-01-16",
  "category": "",
  "debit": "",
  "flags": [
    "is_payroll"
  ]
},
{
  "flags": [
    "is_payroll"
  ],
  "category": "",
  "debit": "",
  "date": "2020-01-02",
  "credit": "820.01",
  "balance": "820.01",
  "details": "Payroll ACME JOB CO"
},
],
"account": "123-12345-1234567",
"score": 1,
"income_type": "monthly_income/paycheck",
"monthly_income": "1899.72",
"future_payments": [
  "2020-02-13",
],
"details": "PAYROLL ACME JOB CO"
}
],
}
],
"status": "Verified"
}
```

Postback Summary

When a Customer has completed their request, you may optionally receive a Postback of the collected data. Please provide an HTTPS URL of your choosing, that accepts a POST with JSON data. The JSON structure is identical to the Fetch API response for your convenience.

Example JSON to receive:

```
{
  "complete_datetime": "2017-06-14 20:32:24",
  "request": "9633F964-513E-11E7-A970-6EC618044181",
}
```

```
"contacts" : [
  {
    "contact" : "johndoe@gmail.com",
    "type" : "email"
  },
  {
    "type" : "phone",
    "contact" : "(604) 555-1234"
  }
],
"accounts" : [
  {
    "bank" : "Inverite",
    "institution" : "001",
    "transit" : "12345",
    "routing_code" : null,
    "account" : "1234567",
    "statistics" : {
      "mean_closing_balance_30" : "221.00",
      "mean_closing_balance" : "163.67"
    },
    "transactions" : [
      {
        "flags" : [
          "is_regular"
        ],
        "category" : "transfer",
        "credit" : "194.71",
        "debit" : "",
        "date" : "2017-06-14",
        "details" : "e-Transfer In Other Reference # 228803",
        "balance" : "217.53"
      }
    ]
  },
  {
    "type" : "chequing",
    "institution" : "123",
    "account" : "1234567",
    "transit" : "12345",
    "bank" : "Test Bank",
    "payschedules" : [
      {
        "frequency" : "bi_weekly:parity_0:thursday",
        "payments" : [
          {
            "details" : "Payroll ACME JOB CO",
            "balance" : "865.82",
            "credit" : "863.92",
            "debit" : "",
            "category" : "",
            "flags" : [
              "is_payroll"
            ],
            "date" : "2020-01-30"
          },
          {
            "details" : "Payroll ACME JOB CO",
            "balance" : "985.49",
            "credit" : "985.30",
            "date" : "2020-01-16",
          }
        ]
      }
    ]
  }
]
```

```
{
  "category": "",
  "debit": "",
  "flags": [
    "is_payroll"
  ],
},
{
  "flags": [
    "is_payroll"
  ],
  "category": "",
  "debit": "",
  "date": "2020-01-02",
  "credit": "820.01",
  "balance": "820.01",
  "details": "Payroll ACME JOB CO"
},
],
"account": "123-12345-1234567",
"score": 1,
"income_type": "monthly_income/paycheck",
"monthly_income": "1899.72",
"future_payments": [
  "2020-02-13",
],
"details": "PAYROLL ACME JOB CO"
}
],
}
],
"status": "Verified"
}
```

Testing Your Own Postback URL

Please test this CURL command set against your own Postback URL. This mimics the exact postback you will receive from our system. Once this has been successfully parsed, we can configure your account to the URL of your choosing.

```
curl -v -H "Accept: application/json" \
  -H "Content-type: application/json" \
  -X POST https://your-postback-url.com/system/inverite_aggregate_postback.php \
  -d '{"request":"9BF8A8C8-47BF-11E7-B37A-F3D4241D77E6","complete_datetime":"2017-06-14
20:32:24","status":"Verified","accounts":[{"institution":"001","transit":"12345","routing_code":null,"account":"1234567","transactio
ns":[{"balance":"217.53","credit":"194.71","details":"e-Transfer In Other Reference #
228803","debit":"","category":"transfer","date":"2017-06-
14","flags":["is_regular"]}],"statistics":{"mean_closing_balance_30":"221.00","mean_closing_balance":"163.67"},"bank":"Inverite"
}]}'
```

```
<?php
$inveriteApiKey = 'my-secret-key';
$inveriteSiteId = '101';
$siteName = 'Brand';
$isDev = 1; // Example code

$response = inveriteCreate(1, '127.0.0.1', 'user@example.com', '6045551313');

$guid = $response['request_guid'];
echo "Store guid $guid to customer database record.";

// wait for postback that verification is complete

$response = inveriteFetch($guid);
$transit = $response['accounts'][0]['transit'];
echo "Store transit $transit to customer database record.";

function inveriteCreate($customerId, $ip, $email = '', $phone = '')
{
    global $inveriteApiKey;
    global $inveriteSiteId;
    global $siteName;
    global $isDev;

    $parameters = array(
        'ip'           => $ip,
        'siteID'       => $inveriteSiteId,
        'referenceid' => $customerId,
        'email'        => $email,
        'phone'        => $phone,
    );

    // in case you want to pass a phone only notification, you must include a unique username parameter
    if ($email == '' && $phone != '')
    {
        $parameters['username'] = $siteName.'_'.$customerId;
    }

    if ($isDev == 1)
    {
        $hostname = 'sandbox.inverite.com';
    }
    else
    {
        $hostname = 'www.inverite.com';
    }
    $ch = curl_init();
    curl_setopt($ch, CURLOPT_URL, "https://$hostname/api/v2/create");
    curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);

    curl_setopt($ch, CURLOPT_HTTPHEADER, array("Content-Type: application/json", "Auth:
$inveriteApiKey"));
    curl_setopt($ch, CURLOPT_POST, 1);
    curl_setopt($ch, CURLOPT_POSTFIELDS, json_encode($parameters));

    $response = curl_exec($ch);

    curl_close($ch);

    return json_decode($response, 1);
}

function inveriteFetch($guid)
{

```

```
global $inveriteApiKey;
global $isDev;

if ($isDev == 1)
{
    $hostname = 'sandbox.inverite.com';
}
else
{
    $hostname = 'www.inverite.com';
}

$ch = curl_init();
curl_setopt($ch, CURLOPT_URL, "https://$hostname/api/v2/fetch/" . $guid);
curl_setopt($ch, CURLOPT_HTTPHEADER, array("Auth: $inveriteApiKey"));
curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);

$response = curl_exec($ch);

curl_close($ch);

return json_decode($response, 1);
}
?>
```

Successful Verification, No Challenge

<u>Username</u>	<u>Password</u>
success	(any)

Successful Verification, Multiple Accounts

<u>Username</u>	<u>Password</u>
success-multiple-accounts	(any)

Failed Verification

<u>Username</u>	<u>Password</u>
error	(any)

Successful Verification with Challenge Question

<u>Username</u>	<u>Password</u>
success-question	(any)

<u>Successful Challenge Answer</u>	(any)
<u>Failed Challenge Answer</u>	wrong

Successful Verification with Challenge Image

<u>Username</u>	<u>Password</u>
success-image	(any)

Successful Verification with Multiple Questions

<u>Username</u>	<u>Password</u>
success-multiple	(any)

Successful Verification with Multi Factor Authentication

<u>Username</u>	<u>Password</u>
success-mfa	(any)